

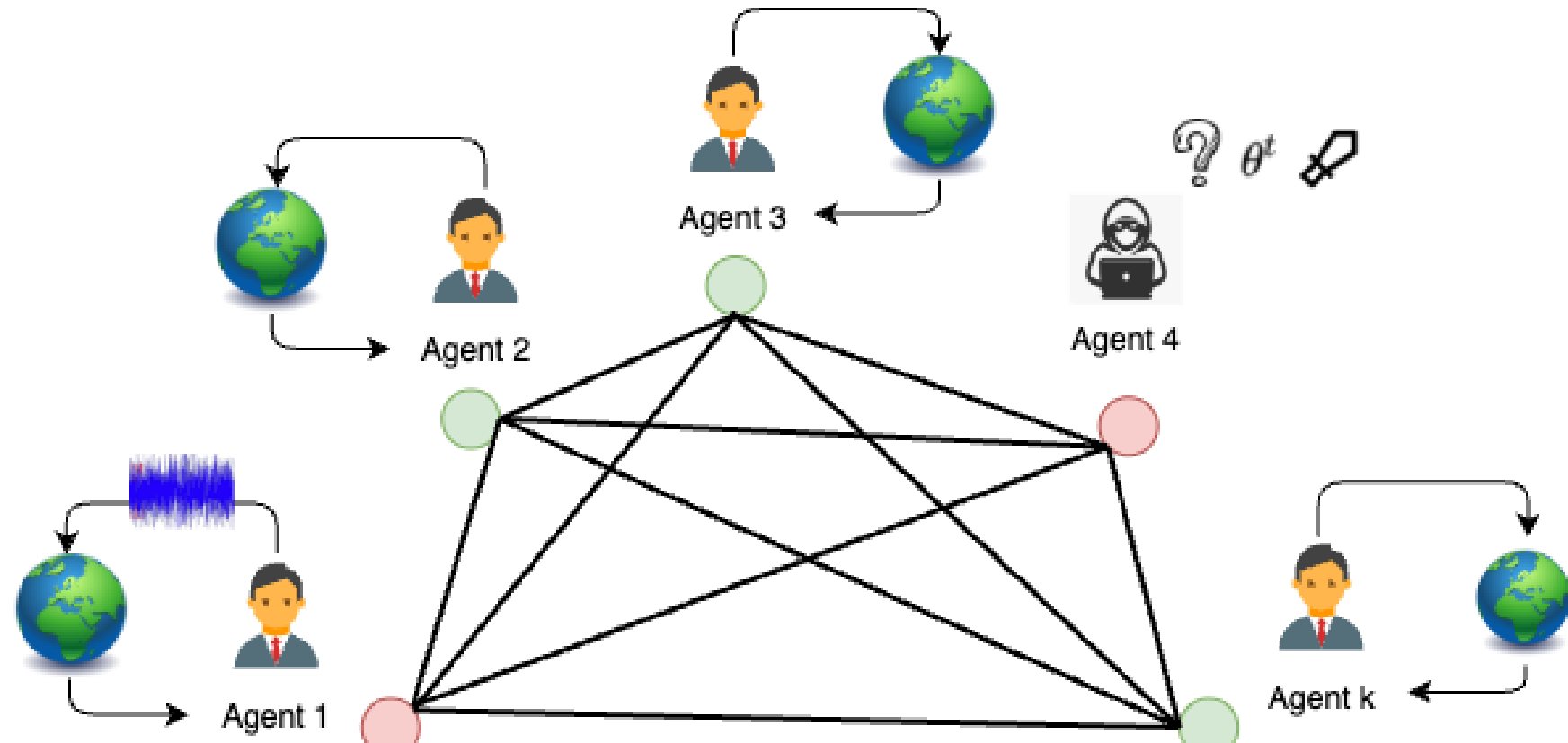
Decentralized Federated Policy Gradient with Byzantine Fault-Tolerance and Provably Fast Convergence



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Motivation

- **decentralized and Byzantine fault-tolerant** federated RL with sample complexity guarantees



- **without** exchange of **raw trajectories** data (due to high cost, e.g., in autonomous driving); only share policy parameters and gradients

Problem Setting

Network setup:

- K agents; **round-based**, synchronous, **all-to-all** communication
- Byz. agents may send arbitrary data as update (may collude!)
- per round $< 1/4$ are Byz., and set of Byz. can change each round (i.e., remembering identities of previously Byz. agents is of no use)

RL setup:

 Each agent $i \in [K]$...

- interacts with a copy of MDP $\mathcal{M} = \{\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma, \rho\}$ (i.e. homogeneous local environments)
- maintains individual policy $\pi_{\theta(i)}$ with local parameters $\theta^{(i)} \in \mathbb{R}^d$
- aims to find $\arg \max_{\theta^{(i)} \in \mathbb{R}^d} J(\theta^{(i)}) := \mathbb{E}_{\tau \sim p(\cdot | \theta^{(i)})} \left[\sum_{t=0}^{H-1} \gamma^t \mathcal{R}(s_t, a_t) \mid \mathcal{M} \right]$

RL assumptions:

 (standard; see, e.g., [1])

- bounded $\|\nabla_{\theta} \log \pi_{\theta}(a \mid s)\|$ and $\|\nabla_{\theta}^2 \log \pi_{\theta}(a \mid s)\|$ over $\theta \in \mathbb{R}^d$
- bounded $\text{Var}[g(\tau \mid \theta)] = \mathbb{E}\|g(\tau \mid \theta) - \nabla J(\theta)\|^2$ over $\theta \in \mathbb{R}^d$
- $\forall \theta_a, \theta_b \in \mathbb{R}^d$, importance weight $\omega(\tau \mid \theta_b, \theta_a) = \frac{p(\tau \mid \theta_a)}{p(\tau \mid \theta_b)}$ well-defined, and $\exists W > 0$ s.t. $\text{Var}[\omega(\tau \mid \theta_b, \theta_a)] \leq W \|\theta_a - \theta_b\|^2$

DEZBYZPG

input: $\theta_0 \in \mathbb{R}^d$, batch sizes N and B , step size η , prob. $p \in (0, 1]$

initialize $\theta_0^{(k)} = \theta_0$

for $t = 0$ **to** $T - 1$ **do**

$c_t \leftarrow \text{Common-Sample}(\text{Be}(p))$

$M \leftarrow \begin{cases} N & \text{if } c_t = 1 \text{ or } t = 0 \\ B & \text{else} \end{cases}$

sample trajectories $\{\tau_{t,i}^{(k)}\}_{i=1}^M$ from $p(\cdot \mid \theta_t^{(k)})$

$$\tilde{v}_t^{(k)} = \begin{cases} \frac{1}{N} \sum_{i=1}^N g(\tau_{t,i}^{(k)} \mid \theta_t^{(k)}) & \text{if } c_t = 1 \text{ or } t = 0 \\ \frac{1}{B} \sum_{i=1}^B g(\tau_{t,i}^{(k)} \mid \theta_t^{(k)}) + \frac{1}{\eta} (\theta_t^{(k)} - \theta_{t-1}^{(k)}) - \frac{1}{B} \sum_{i=1}^B g^{\omega_{\theta_t^{(k)}}}(\tau_{t,i}^{(k)} \mid \theta_{t-1}^{(k)}) & \text{else} \end{cases}$$

$v_t^{(k)} \leftarrow \text{Aggregate}(\langle \tilde{v}_t^{(k')} \rangle_{k'=1}^K)$

$\tilde{\theta}_{t+1}^{(k)} = \theta_t^{(k)} + \eta v_t^{(k)}$

$\theta_{t+1}^{(k)} \leftarrow \text{Avg-Agree}_{\kappa}(\langle \tilde{\theta}_{t+1}^{(k')} \rangle_{k'=1}^K)$

output: $\theta_{\hat{T}}^{(k)}$ with $\hat{T} \sim \text{Common-Sample}(\mathcal{U}([T]))$

Aggregation & Agreement

Robust Aggregation: $\exists C_{ra} > 0$ s.t. $\forall \alpha \in [0, 1/4)$, any $\langle \theta^{(k)} \rangle_{k \in [K]}$, any $\mathcal{H} \subseteq K$ with $|\mathcal{H}| \geq (1 - \alpha)K$, and $\hat{\theta} := \text{Aggregate}(\theta^{(1)}, \dots, \theta^{(K)})$,

$$\mathbb{E} [\|\hat{\theta} - \bar{\theta}\|^2] \leq \frac{C_{ra} \alpha}{|\mathcal{H}|(|\mathcal{H}| - 1)} \sum_{i, l \in \mathcal{H}} \mathbb{E} [\|\theta^{(i)} - \theta^{(l)}\|^2].$$

Averaging Agreement: similarly, $\exists C_{avg} > 0$ s.t. running **Avg-Agree** $_{\kappa}$ on any inputs $\langle \theta^{(k)} \rangle_{k \in [K]}$ for $\kappa \in \mathbb{N}$ rounds, outputs $\langle \hat{\theta}^{(k)} \rangle_{k \in [K]}$ s.t.

$$\max_{i, l \in \mathcal{H}} \|\hat{\theta}^{(i)} - \hat{\theta}^{(l)}\| \leq \frac{\max_{i, l \in \mathcal{H}} \|\theta^{(i)} - \theta^{(l)}\|}{2^{\kappa}} \quad \text{and} \quad \|\bar{\hat{\theta}} - \bar{\theta}\| \leq C_{avg} \cdot \max_{i, l \in \mathcal{H}} \|\theta^{(i)} - \theta^{(l)}\|.$$

Convergence Results

- **K -agent α -tolerant ϵ -approximate solution:**

For $\mathcal{G}_T \subset [K]$ honest agents at round T , we want $\forall k \in \mathcal{G}_T$, $\mathbb{E}[\|\nabla J(\theta_T^{(k)})\|^2] \leq \epsilon^2$ where $\theta_T^{(k)}$ is agent k 's output.

- **Main result:** For suitably chosen η, κ, N, B, p , the expected sample complexity for DEZBYZPG to achieve a K -agent α -tolerant ϵ -approximate solution is

$$\mathcal{O} \left(\frac{\alpha^{3/2}}{\epsilon^4} + \frac{\alpha^{1/2}}{K \epsilon^4} + \frac{\alpha^{1/2}}{K^{1/2} \epsilon^3} + \frac{1}{K \epsilon^3} \right).$$

$$\alpha = 0 \Rightarrow \mathcal{O}(K^{-1} \epsilon^{-3}); \quad \alpha = \mathcal{O}(K^{-1}) \Rightarrow \mathcal{O}(K^{-3/2} \epsilon^{-4} + K^{-1} \epsilon^{-3})$$

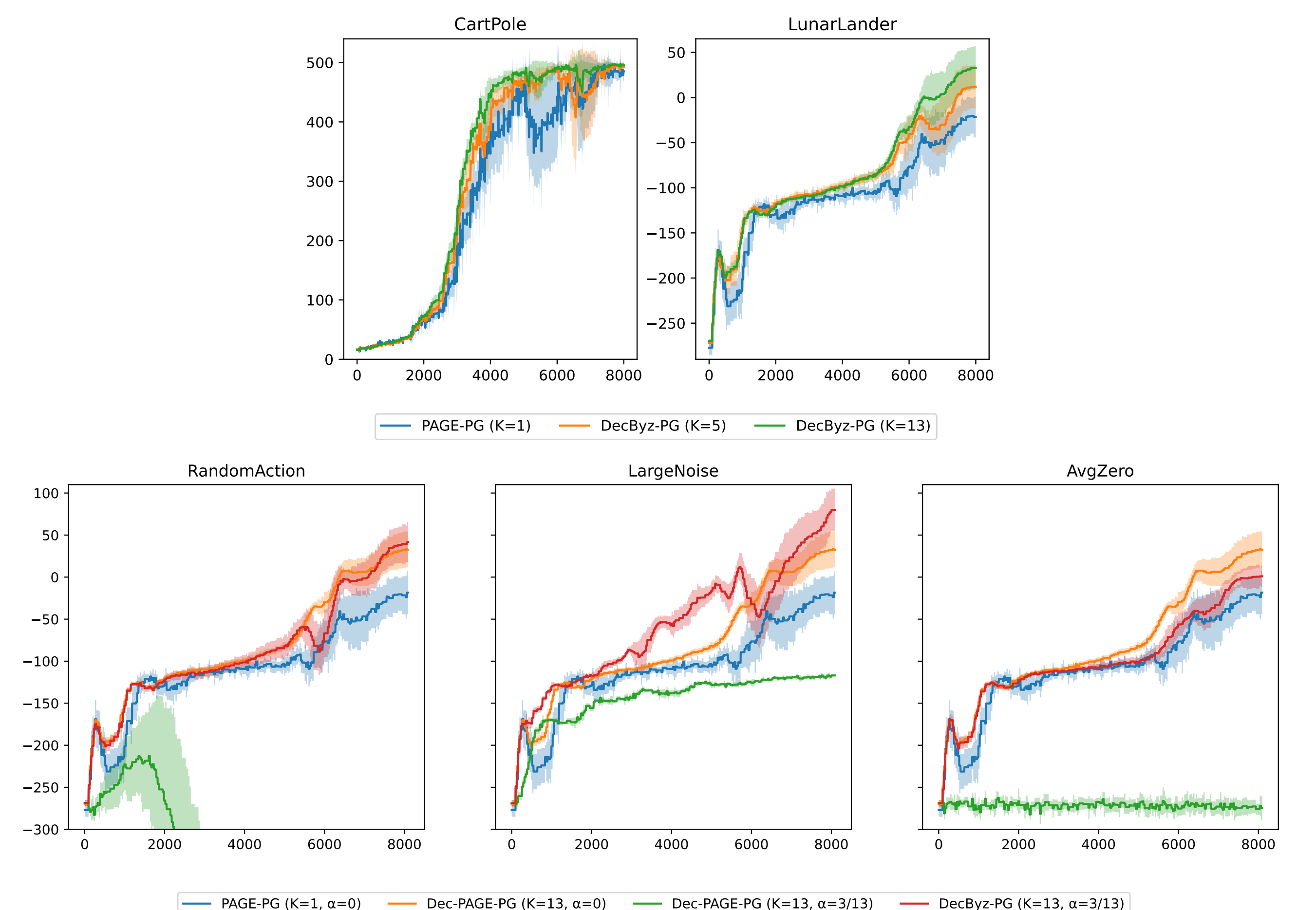
- **In comparison:** best non-Byz.-tol. $\mathcal{O}(K^{-1} \epsilon^{-3})$, vanilla PG $\mathcal{O}(\epsilon^{-4})$

Experiments

Attack types:

 Byzantine agents ...

- **RandomAction:** ... sample action u.a.r. instead of from $\pi_{\theta}(\cdot \mid s)$
- **LargeNoise:** ... send large-variance noise instead of $\tilde{v}_t^{(k)} / \tilde{\theta}_{t+1}^{(k)}$
- **AvgZero:** ... collude to choose gradients s.t. overall mean is 0



Future Work

- tolerating more than 1/4-fraction of Byzantine agents possible?
- heterogeneous environments (effect on fault-tol. & convergence)
- decentralized methods with fault-tolerance and privacy guarantees

References

- [1] Pan Xu, Felicia Gao, and Quanquan Gu. An improved convergence analysis of stochastic variance-reduced policy gradient. In *Uncertainty in Artificial Intelligence*, pages 541–551. PMLR, 2020.